MODULE 1

INTRODUCTION TO DATA SCIENCE

Definitions • Skills • Tools • Data Types • Collections • Preprocessing • Analysis & Analytics

Study Notes

This module introduces the basic concepts of Data Science, the skills and tools required by a data scientist, common data types and collections, data preprocessing, and the major types of data analysis.

1. Definition of Data Science

Data Science is an interdisciplinary field that uses statistics, mathematics, programming, data analysis, machine learning, and domain knowledge to extract useful information and insights from data.

Key idea: Data Science converts raw data into meaningful insights that can support decisions, predictions, and problem solving.

Typical Data Science workflow:

- Collect data from databases, files, sensors, websites, applications, or APIs.
- Clean and preprocess the collected data.
- Explore and analyze the data.
- Build statistical or machine-learning models when required.
- Evaluate the results.
- Communicate insights using reports, dashboards, or visualizations.

2. Skills Required for Data Science

A data scientist needs a combination of technical, analytical, and communication skills.

Technical skills

- Programming – Python, R, Java, or other languages.
- Statistics and probability – distributions, hypothesis testing, correlation, regression, etc.
- Mathematics – linear algebra, calculus, and optimization basics.
- Data handling – cleaning, transformation, and database operations.
- Machine Learning – supervised and unsupervised learning concepts.
- Data visualization – charts, dashboards, and visual storytelling.

Non-technical skills

- Problem-solving and critical thinking.
- Domain knowledge.
- Communication and presentation.
- Business understanding and decision-making.

3. Tools for Data Science

Data scientists use different tools for programming, data storage, analysis, visualization, and machine learning.

Category	Examples	Purpose
Programming	Python, R, SQL	Programming, querying and data manipulation
Development	Jupyter Notebook, VS Code	Writing and running analysis/code
Python libraries	NumPy, Pandas, Matplotlib, Seaborn	Numerical work, data handling and visualization
Machine Learning	Scikit-learn, TensorFlow, PyTorch	Model development
Databases	MySQL, PostgreSQL, MongoDB	Data storage and retrieval
Visualization	Power BI, Tableau	Dashboards and business reporting

4. Data Types

A data type describes the kind of value stored in a variable or dataset column.

Common data types

- Numerical: numbers used for measurements or quantities. Example: age, salary, height.
- Integer: whole numbers. Example: 10, 25, 100.
- Float: decimal numbers. Example: 72.5, 3.14.
- Categorical: values representing groups or categories. Example: Department = HR, Sales, IT.
- Binary: two possible values, such as Yes/No or 0/1.
- Ordinal: categories with an order. Example: Low, Medium, High.
- Text/String: words or sentences. Example: customer comments.
- Date/Time: dates and timestamps. Example: 2026-10-06.

Important: Correctly identifying data types is necessary before analysis because different types require different operations.

5. Data Collections

Data collections are structures used to store and organize multiple values. In Python, the main built-in collections are list, tuple, set, and dictionary.

Collection	Characteristics	Example
List	Ordered, changeable, allows duplicates	[10, 20, 20, 30]
Tuple	Ordered, generally immutable, allows duplicates	(10, 20, 30)
Set	Unordered collection of unique values	{10, 20, 30}
Dictionary	Key-value pairs	{'name': 'Ravi', 'age': 20}

6. Data Preprocessing

Data preprocessing is the process of preparing raw data so that it becomes suitable for analysis and machine-learning tasks.

Main preprocessing steps

- Data cleaning – handle missing values, duplicates, incorrect values, and inconsistent formats.

- Data integration – combine data from multiple sources.
- Data transformation – change data into a suitable form.
- Encoding – convert categorical values into numerical representations when required.
- Feature scaling – bring numerical features to comparable scales using methods such as normalization or standardization.
- Outlier handling – identify and appropriately treat unusual observations.
- Feature selection – retain relevant variables and remove unnecessary ones.

Example: A salary column containing ₹50,000, "50000", and missing values needs cleaning and conversion before reliable numerical analysis.

7. Data Analysis and Data Analytics

Data analysis is the process of inspecting, cleaning, transforming, and interpreting data to discover useful information. Data analytics is the broader practice of using data, analytical methods, tools, and models to answer questions and support decisions.

Data Analysis

- Focuses on examining data and finding patterns, relationships, trends, and abnormalities.

Data Analytics

- Uses analysis along with statistical, computational, and business techniques to generate actionable insights.

Example: An analyst may analyze monthly sales data to discover that sales are falling. Analytics can go further by identifying likely causes and recommending actions.

8. Descriptive Analysis

Descriptive analysis answers: “What happened?” It summarizes historical data using numbers, tables, and visualizations.

- Mean, median, mode, minimum, maximum, and standard deviation.
- Frequency tables and percentages.
- Bar charts, line charts, histograms, and dashboards.

Example: “The company sold 12,000 units in September, which was 8% higher than August.”

9. Diagnostic Analysis

Diagnostic analysis answers: “Why did it happen?” It investigates relationships, patterns, and possible causes behind an observed result.

- Compare groups or time periods.
- Drill down into detailed data.
- Examine correlations and relationships.
- Identify factors associated with an outcome.

Example: If September sales increased, diagnostic analysis may determine that a festival promotion and discount campaign contributed to the increase.

10. Predictive Analysis

Predictive analysis answers: “What is likely to happen?” It uses historical data, statistics, and machine-learning models to estimate future or unknown outcomes.

- Regression can predict numerical values such as sales or price.
- Classification can predict categories such as churn/not churn.
- Time-series methods can forecast future values.

Example: A company predicts which customers are likely to stop using its service next month.

11. Prescriptive Analysis

Prescriptive analysis answers: “What should we do?” It uses analytical results, constraints, and sometimes optimization or simulation to recommend actions.

- Suggests possible decisions or actions.
- Can compare alternative strategies.
- May use optimization, simulation, rules, or predictive-model outputs.

Example: If demand is predicted to increase, prescriptive analysis may recommend increasing inventory by a suitable amount.

12. Exploratory Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of examining a dataset to understand its structure, distributions, relationships, missing values, outliers, and important patterns before formal modeling.

Common EDA activities

- Inspect rows, columns, data types, and summary statistics.
- Check missing values and duplicates.
- Study distributions using histograms and density plots.
- Study relationships using scatter plots, correlation matrices, and grouped summaries.
- Identify outliers and unusual observations.
- Generate hypotheses for further analysis.

Example: Before building a salary-prediction model, EDA can reveal missing salaries, unusual salary values, department differences, and relationships with experience.

13. Mechanistic Analysis

Mechanistic analysis focuses on understanding the underlying mechanism or physical/process relationships that explain how inputs produce an outcome. It is more concerned with the structure of the process than simply finding statistical association.

- Attempts to explain how a system works.
- Uses domain knowledge, mathematical relationships, or physical/process rules.
- Can be useful when cause-and-effect mechanisms are known or can be modeled.

Example: In an engineering system, mechanistic analysis may model how temperature, pressure, and flow rate interact to determine system behavior.

14. Types of Analysis – Quick Comparison

Type	Main Question	Purpose
Descriptive	What happened?	Summarize past data

Type	Main Question	Purpose
Diagnostic	Why did it happen?	Find causes and relationships
Predictive	What is likely to happen?	Forecast future/unknown outcomes
Prescriptive	What should we do?	Recommend actions
Exploratory	What patterns exist?	Discover patterns and generate hypotheses
Mechanistic	How does the system work?	Model underlying mechanisms

15. Important Exam Points

- Data Science: combines programming, statistics, mathematics, domain knowledge, and analytical methods to extract useful insights from data.
- Data preprocessing: prepares raw data by cleaning, transforming, integrating, encoding, scaling, and handling outliers.
- Descriptive: What happened?
- Diagnostic: Why did it happen?
- Predictive: What is likely to happen?
- Prescriptive: What should we do?
- Exploratory: Discover patterns and relationships in data.
- Mechanistic: Understand the underlying mechanism of a system.